

CI/CD Report – HW06 API Testing

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Pipeline: `.github/workflows/api-tests.yml`
Repository: <https://github.com/huynhgiaau27112005/23127153-hw06-ai-api>

Pipeline Overview

```
```mermaid
flowchart LR
 A[Push / PR / workflow_dispatch] --> B[Checkout homework]
 B --> C[Checkout ttbhanh/eshop-sut]
 C --> D[Setup Node 20]
 D --> E[npm ci]
 E --> F[build-postman-from-cases.js]
 F --> G[Start EShop on :3010]
 G --> H[Newman run]
 H --> I[Upload HTML artifact]
```
```

Triggers

| Event | Condition |
|-------------------|---|
| push | main or master, paths: test-cases, postman, scripts, workflow |
| pull_request | Target main or master |
| workflow_dispatch | Manual with fail_one_test input (true/false) |

Workflow Inputs

| Input | Type | Default | Effect |
|---------------|--------|---------|--|
| fail_one_test | choice | false | When true, adds --bail to Newman (stop on first failure) |

Two Sample Runs (required)

| # | Input | Run URL | Conclusion |
|---|----------------------------------|---|---|
| 1 | fail_one_test=false (full suite) | https://github.com/huynhgiaau27112005/23127153-hw06-ai-api/actions/runs/33755055651 | Job completed; Newman exit ≠ 0 due to known SUT bugs (documented). Artifact newman-report uploaded. |
| 2 | fail_one_test=true (--bail) | https://github.com/huynhgiaau27112005/23127153-hw06-ai-api/actions/runs/33755064475 | Job completed; stops earlier on first failure. Artifact uploaded. |

Screenshots of these runs: open the Actions URLs above (workflow_dispatch details show the input values). CI HTML report is available as the newman-report artifact on each run.

Steps Detail

1. ****Checkout homework**** - `actions/checkout@v4`
2. ****Checkout SUT**** - `ttbhanh/eshop-sut` into `eshop-sut/`
3. ****Node setup**** - v20 with npm cache
4. ****Install**** - `npm ci` fallback to `npm install`
5. ****Build collection**** - `node scripts/build-postman-from-cases.js`
6. ****Start SUT**** - `PORT=3010 node server.js` in `eshop-sut/backend`
(wait loop for `/api/products`)
7. ****Newman**** - full collection, `cli` + `htmlextra` reporters
8. ****Artifact**** - uploads `reports/newman-ci.html` (always)

Local Equivalent

```
`powershell
.\scripts\run-newman.ps1
.\scripts\run-newman.ps1 -FailOneTest
.\scripts\run-newman.ps1 -Folder "FR-02 Login"
`
```

Known CI Limitations

1. State-dependent tests (login logout, deletes) may fail in a full sequential suite
2. `continue-on-error` on Newman keeps the workflow green so artifacts always upload for grading
3. Some assertions document real SUT defects (see `docs/bugs.md` + GitHub Issues #1-#6)

Recommendations

- Split pools into parallel matrix jobs
- Fail the job on Newman exit code after artifact upload (stricter gate)
- Seed dedicated users per CI run to avoid lockout interference